

Xcos: Simulate Hybrid Systems Easily

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A common problem facing a lot of independent designers is the choice of tools they have for their simulation needs. "Do I need precise and fast help that a paid licence from MATLAB/Simulink offers, or can I work with a tool that has documentation mostly offered by its users?"

As with most answers in life, this one too is an, "It depends." The final decision depends on the scope of the project, time limit and, may be, the depth of your pockets. This month we introduce you to a little simulator by the name of Xcos, developed by Scilab Enterprises.

What is Xcos and how it can help

Numerical simulation is very important for all designers, almost indispensable to the system design process. Instead of wasting money on real experiments like someone from 1980s, today it is all about simulations and their results. Complex phenomena like electrical and mechanical interactions are modelled at the smallest level, and today's engineers know that they need to be aware of these techniques in order to be better than good at what they do.

Distributed along with Scilab, Xcos is a tool used to model and simulate hybrid dynamic systems. Users can model discrete as well

as continuous and causal, as well as acausal, models on it along with being able to design, load, save, compile and simulate. Most commonly-used blocks, namely, signal processing, electrical systems, thermo-hydraulic systems, mathematical operations and much more, are available for use on Xcos. There are ways to import user-defined blocks that can be easily added to Xcos framework, which is flexible

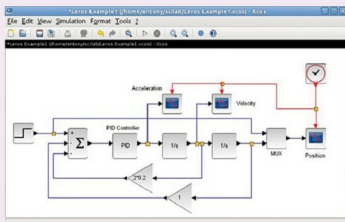


Fig. 1: Typical example of Xcos

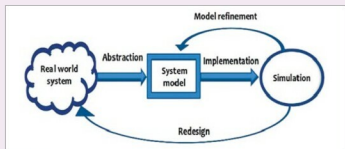


Fig. 2: Role of simulation

Software at a glance

Developer	: Scilab Enterprises
Latest version	: 5.5.1 October 2, 2014
Written in	: Scilab, Java, C and FORTRAN
Environment	: Linux, Windows, Mac OS and BSD

enough to handle all that.

Association of Scilab with LabView allows it access to numerous free gateways and links, which is critical for enhanced performance. It also connects to National Instruments' NIDAQ module for better access to data-acquisition I/O devices.

Like other simulation, different blocks represent different functions, pre-defined or user-defined. Users can arrange these in order of requirement of the system and run the simulation to see the effects. Xcos is freely distributed under CeCILL v2 licence, and can be run on Windows, Linux and Mac OS X.

General environment

After launching Xcos, you can see two windows, namely, a palette browser, which houses all pre-defined blocks and an editing window where you can place the blocks and make requisite connections. To design a circuit, you can simply pick these up from the palette browser and drop it into the editing window. And, connect the blocks using their different ports, inputs, outputs and events so as to simulate the given circuit.

Features of Xcos

Xcos environment supports simulation as well as modelling, and its systems include the following features: